

Symphonym: Universal Phonetic Embeddings for Cross-Script Name Matching

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Abstract

Linking names across historical sources, languages, and writing systems remains a fundamental challenge in digital humanities and geographic information retrieval. Existing approaches require language-specific phonetic algorithms or fail to capture phonetic relationships across different scripts. This paper presents Symphonium, a neural embedding system that maps names from any script into a unified 128-dimensional phonetic space, enabling direct similarity comparison without runtime phonetic conversion. Symphonium uses a Teacher-Student architecture where a Teacher network trained on articulatory phonetic features produces target embeddings, while a Student network learns to approximate these embeddings directly from characters. The Teacher combines Epitran (extended with 100 new language-script mappings), Phonikud for Hebrew, and CharsiuG2P for Chinese, Japanese, and Korean. Training used 29.9 million triplet samples of toponyms spanning 20 writing systems from GeoNames, Wikidata, and Getty Thesaurus of Geographic Names. On the MEHDIE Hebrew-Arabic historical toponym benchmark, Symphonium achieves Recall@10 of 97.6% and MRR of 90.3%, outperforming Levenshtein and Jaro-Winkler baselines (Recall@1: 86.7% vs 81.5% and 78.5%). Evaluation on 12,947 real cross-script training pairs shows 82.6% achieve ≥ 0.75 cosine similarity, with best performance on Arabic-Cyrillic (94–100%) and Cyrillic-Latin (94.3%) combinations. The fixed-length embeddings enable efficient retrieval in digital humanities workflows, with a case study on medieval personal names demonstrating effective transfer from modern place names to historical orthographic variation.

Keywords: phonetic embeddings; cross-script matching; teacher-student distillation; neural networks; digital humanities; name matching

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1 Introduction

Historical gazetteers aggregate place references from sources spanning millennia, dozens of languages, and multiple writing systems. The World Historical Gazetteer¹ (WHG) alone indexes over 67 million toponym records from antiquity to the present, drawn from sources as diverse as classical Latin itineraries, medieval Arabic geographies, and modern national mapping agencies. Each toponym record is tagged with language and script metadata, though the same surface form (e.g., “London”) may appear multiple times when attested in different languages.

The core difficulty is that place name similarity is fundamentally *phonetic*, not orthographic. English speakers recognise “København” and “Copenhagen” as the same city not because the spellings match, but because the sounds are similar enough to suggest common origin. Yet computational approaches to toponym matching have largely relied on string-based metrics (edit distance, Jaro-Winkler) or phonetic algorithms designed for single languages (Soundex, Metaphone for English; Cologne phonetic for German). These approaches fail catastrophically when names cross script boundaries: no amount of edit distance tuning will link “東京” to “Tokyo”.

Recent work has applied neural embeddings to geographic entity matching (Qiu et al., 2024; Rama, 2016; Sagi et al., 2025), but existing systems either operate within single scripts, require language identification at inference time, or depend on curated transliteration resources that don’t scale to the long tail of historical languages. Until now, there has been no system that can place “Москва”, “Moscow”, “موسكو”, and “モスクワ” (Japanese Katakana) near each other in embedding space using only the raw character sequences as input.

This paper presents such a system, which I call *Symphonym*. The key contributions are:

1. A **script-aware but script-agnostic embedding architecture** that maps toponyms from 20+ writing systems into a unified 128-dimensional space where proximity reflects phonetic similarity.
2. A **Teacher-Student training framework** that transfers phonetic knowledge from articulatory features derived from IPA transcriptions (via Epi-tran, Phonikud, and CharsiuG2P) to a character-level model requiring no phonetic resources at inference time.

¹<https://whgazetteer.org>

3. A **noise-aware training regime** with strategic language dropout that forces the model to learn script-intrinsic phonetic patterns rather than relying on language-specific shortcuts.
4. **Phonetic similarity filtering** for training pairs using PanPhon feature distance to prevent the model from learning false equivalences between unrelated exonyms (e.g., “Germany” $\neq$ “Deutschland”). This filtering ensures training samples represent genuine phonetic correspondences—variants that sound alike—rather than semantic or cultural name substitutions. The filtering improves model precision without biasing toward particular scripts or languages, as it operates on universal articulatory feature space.
5. A **three-phase curriculum** progressing from phonetic feature learning through knowledge distillation to hard negative discrimination.
6. **Empirical validation** of cross-domain transfer: a case study on medieval personal name matching (Section 8) demonstrates that embeddings trained on modern toponyms transfer effectively to historical personal names with pre-standardisation orthography.

The resulting system will enable fuzzy phonetic search across the full WHG corpus without language identification, script-specific preprocessing, or runtime phonetic conversion. Beyond cross-script matching, the phonetic embedding approach naturally handles *historical orthographic variation*—the inconsistent spellings characteristic of pre-standardisation documents—by grouping sound-alike variants near their modern canonical forms. This capability extends beyond toponyms to personal names, and in any textual domain involving phonetically consistent but orthographically variable name forms.

Scope and Integration. This work addresses the *name-matching* component of toponym resolution specifically. The model has no access to geographic coordinates or spatial context—it operates purely on phonetic similarity between strings. Both Symphonium and WHG’s new reconciliation pipeline (termed WHG PLACE: Place Linkage, Alignment, and Concordance Engine) build on architecture developed by Grossner and Mostern (2021). Within PLACE, phonetic similarity serves as one input to a larger process: candidate toponyms are first retrieved via the phonetic-aware toponym index, then places containing matching toponyms are selected from the places index and filtered by geographic proximity and other contextual criteria. The contribution here is the phonetic similarity component alone.

In practical terms, Symphonium will enhance WHG’s Reconciliation Service API² (v2.0), facilitating the work of digital humanities and GLAM (galleries, libraries, archives, museums) professionals who seek to link place references in

²<https://docs.whgazetteer.org/content/technical/apis.html#reconciliation-service-api>

historical documents, archival catalogues, and research datasets to WHG’s consolidated index. Once linked, these references become entry points to a web of related materials—other sources attesting the same places, variant historical names, and cross-references across collections—even when the original materials span multiple scripts, languages, and historical periods.

2 Related Work

Classical and String-based Approaches. Early phonetic matching algorithms—Soundex (Russell, 1918), Metaphone (Philips, 1990), and PHONIX (Gadd, 1990)—rely on hand-crafted rules for specific languages and scripts. PHONIX exemplifies the limits of manual engineering: Gadd encoded 160 transformation rules for Latin-script names in Southern African languages, whereas Symphonym learns phonetic correspondences from 29.9 million samples across 20 writing systems. Edit distance metrics (Levenshtein, Jaro-Winkler) measure orthographic rather than phonetic similarity and fail across scripts: Levenshtein distance treats “Moscow” and “Москва” as maximally dissimilar (sharing no characters) despite very similar pronunciation. While Recchia and Louwerse (Recchia and Louwerse, 2013) and Santos et al. (Santos et al., 2018) demonstrate that combining multiple string metrics improves within-script matching, these approaches cannot generalise to cross-script scenarios.

Neural Entity Matching. Recent neural approaches to geographic entity resolution operate within constrained scopes. Qiu et al. (Qiu et al., 2024) achieve strong performance on Chinese address matching using BERT pre-trained on geographic text combined with Chinese-specific features (Pinyin, radicals, stroke counts), but these features lack analogues in other scripts. The MEHDIE project (Sagi et al., 2025) addresses Hebrew-Arabic historical toponym matching, demonstrating that direct script-to-script approaches can outperform romanisation for related languages, though such methods require manual rules for each language pair. Rama (Rama, 2016) applies Siamese neural networks to cognate identification, learning sound correspondences from character-level representations. A key limitation of purely data-driven approaches is that they must discover phonetic relationships (e.g., that ‘b’ and ‘p’ share articulatory properties) through co-occurrence patterns alone, whereas Symphonym’s use of PanPhon articulatory features explicitly encodes these relationships, enabling more efficient learning and better generalisation to low-resource scripts.

Phonetic Grounding and Architecture. The PWESUITE evaluation by Zouhar et al. (2024) provides systematic comparison of phonetic word embedding methods. Their findings demonstrate that models incorporating articulatory features show improved cross-language transfer compared to approaches using only IPA or character-level inputs, and that phonetically grounded models trained on one

language can generalise to others. These findings directly inform my Teacher network design, which builds on articulatory feature representations. However, PWESUITE addresses monolingual embeddings within single scripts; my contribution extends validated phonetic grounding techniques to the multi-script setting via Teacher-Student distillation. Siamese networks (Bromley et al., 1993; Chopra et al., 2005) provide the architectural foundation for similarity learning, with applications ranging from face verification (Schroff et al., 2015) to text matching (Mueller and Thyagarajan, 2016). I adopt cosine similarity rather than Manhattan distance (L_1) to avoid length penalties and better capture directional phonetic relationships (Lin et al., 2017).

Cross-lingual Representations. The broader cross-lingual embedding literature (Conneau et al., 2018; Artetxe et al., 2018) demonstrates techniques for aligning monolingual semantic spaces, but targets word meaning rather than phonetic form. The TRANSLIT resource (Benites et al., 2020) provides a large-scale collection of name variants across numerous languages for training transliteration systems, capturing conventionalised mapping standards. However, Symphonym’s native script processing—CJK via CharsiuG2P (Zhu et al., 2022), Hebrew via Phonikud (Kolani et al., 2025), Cyrillic/Arabic/Greek via extended Epi-tran (Mortensen et al., 2018)—sidesteps romanisation variance by operating in IPA-derived phonetic space. PanPhon (Mortensen et al., 2016) maps IPA to articulatory feature vectors, forming the Teacher network’s phonetic foundation.

3 Architecture

My system employs a Teacher-Student architecture where a Teacher network, trained on articulatory phonetic features, produces target embeddings that a Student network learns to approximate from character sequences alone. At inference time, only the Student is required, enabling deployment without phonetic resources. This design addresses the limitations of prior approaches discussed in Section 2, particularly the reliance on language-specific phonetic rules and single-script operation.

3.1 Design Principles

Three principles guide my architecture:

Script Awareness Without Script Dependence. The model must handle 20 writing systems but produce embeddings in a unified space where script boundaries are transparent. I achieve this through deterministic script detection (via Unicode block analysis) combined with learned script embeddings that allow the model to interpret characters appropriately for each script.

Phonetic Grounding. Embedding similarity must reflect phonetic similarity, not orthographic or semantic similarity. I ground the embedding space through the Teacher network, which operates on articulatory features derived from IPA transcriptions.

Inference Efficiency. The deployed model must encode toponyms without runtime phonetic conversion, language identification, or external resources. All phonetic knowledge is distilled into the Student’s parameters during training.

3.2 Architecture Overview

Figure 1 illustrates the Teacher-Student architecture and the three-phase training curriculum.

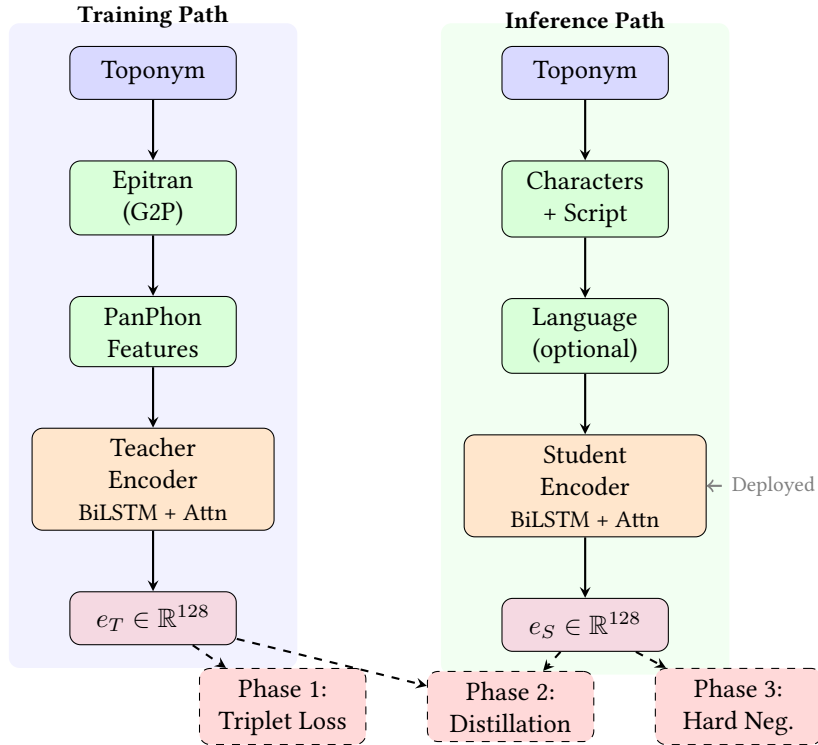


Figure 1: Teacher-Student architecture. **Left:** The Teacher pathway converts toponyms to IPA via Epitrans, extracts PanPhon articulatory features, and encodes to 128-d embeddings. Used only during training. **Right:** The Student pathway processes raw characters with script/language metadata. Deployed at inference time without phonetic resources. **Bottom:** Three training phases progressively transfer phonetic knowledge from Teacher to Student.

3.3 Script Detection and Character Vocabulary

I detect script from Unicode code points, mapping each character to one of 20 script categories. Analysis of the 66.9 million toponyms reveals 8,448 unique observed characters. The vocabulary contains **113,309 tokens** spanning alphabetic scripts (with expansion to full Unicode ranges), approximately 93,500 observed Han characters from the CJK Unified Ideographs block, and 11,624 Hangul syllables. Japanese Kanji are treated as CJK characters, while Hiragana and Katakana are classified as separate script categories. Korean Hangul syllables are decomposed into 51 Jamo components for the Student network’s character embeddings, providing explicit phonetic structure. For IPA transcription, Chinese topolects (Mandarin, Gan, Wu, Yue) and Korean are processed via CharsiuG2P (Zhu et al., 2022), and Hebrew via Phonikud (Kolani et al., 2025). Japanese presents a routing challenge: v6 incorrectly routes all `lang=ja` toponyms to CharsiuG2P (which only processes Kanji), despite native Epitran support via `jpn-Hira` and `jpn-Kana` codes (see §9). Each token is tagged with its source script, and the vocabulary supports 1,944 distinct language codes observed in the corpus.

3.4 Teacher Network: Phonetic Feature Encoder

The Teacher network encodes toponyms via IPA transcriptions and articulatory features (Figure 1, left). Given a toponym, IPA transcriptions are obtained through appropriate G2P methods for each script, then PanPhon features (Mortensen et al., 2016) are extracted—a 24-dimensional binary vector per phoneme encoding articulatory properties (place, manner, voicing). The architecture comprises: feature projection to hidden dimension, bidirectional LSTM, multi-head self-attention, learned attention pooling, and output projection to 128 dimensions with L2 normalisation. The Teacher is trained on triplets of co-located toponyms, learning to place phonetically similar names (same place, different languages) closer than dissimilar names (different places).

3.5 Generalisation Through Articulatory Features

Grounding the embedding space in PanPhon articulatory features enables generalisation to languages with limited training representation. PanPhon represents each IPA segment as a 24-dimensional binary vector encoding universal articulatory properties (place, manner, voicing), independent of language identity. The phoneme /b/ receives identical features whether appearing in English “Berlin”, Russian “Берлин”, or Arabic “برلين”—all encode a voiced bilabial stop. When the Teacher learns that toponyms with similar PanPhon feature sequences should have similar embeddings, it learns relationships between articulatory configurations, not specific languages. This knowledge transfers automatically to any script representing the same sounds, even those absent from training. For scripts with limited Student coverage, the Teacher network can be used directly at inference if IPA transcription is available—valuable for historical or rare scripts.

3.6 Student Network: Universal Character Encoder

The Student network processes raw character sequences with script and language metadata (Figure 1, right). Each character maps to a learned 96-dimensional embedding via script-partitioned vocabulary lookup. Script (deterministically detected) and language (ISO 639 code when available, otherwise <UNK>) contribute 16-dimensional embeddings concatenated to character embeddings. During training, known languages are randomly replaced with <UNK> (50% probability), forcing the model to learn script-intrinsic patterns. The architecture mirrors the Teacher: BiLSTM, self-attention, attention pooling, and output projection to 128 dimensions with L2 normalisation.

3.7 Noise Augmentation

To prepare the model for handling OCR errors, historical spelling variations, and transcription inconsistencies, I apply character-level noise during Student training (Phases 2 and 3): insertions, deletions (probability 0.1 each), substitutions and transpositions (probability 0.05 each), using characters from the same script. Noise is applied with 30% probability per training example, with the target being the Teacher’s embedding of the *clean* input, training the Student to map corrupted inputs to the same phonetic region as their clean counterparts.

4 Training Data

4.1 Data Selection and Curation Criteria

I extract training data from WHG’s `places` index (geographic entities) and `toponyms` index (name variants with script/language metadata), restricting to three major namespace authorities—GeoNames (`gn`), Wikidata (`wd`), and the Getty Thesaurus of Geographic Names (`tgn`)—while excluding less rigorously-curated sources. Wikidata inclusion is critical for non-Latin script coverage, particularly South Asian (Devanagari, Tamil, Telugu, Kannada, Malayalam, Bengali) and Middle Eastern (Arabic, Hebrew) scripts where GeoNames and TGN lack sufficient multilingual correspondences. Training pairs are generated from co-located name attestations: when an authority lists multiple variants (e.g., “London”, “Londres”, “Лондон”) within the same place record, I construct triplets as described in Section 3. To address script and language imbalances, I define four selection criteria:

1. Script-Language Stratified Sampling. To prevent dominance by Latin script and European languages, I implement quota-based sampling by script+language pair (e.g., LATIN:en, CYRILLIC:ru), ensuring representation of under-resourced languages within well-resourced scripts. Each bin is capped at 50,000 samples; bins below 1,000 are dropped, while small bins (1,000–50,000) may be oversampled up to $5\times$. This prevents high-resource bins from dominating gradient signal while ensuring coverage across all script-language combinations.

2. Global Vocabulary Construction. To prevent out-of-vocabulary errors at inference, I decouple vocabulary construction from pair generation. I scan the entire corpus (66.9M unique toponyms) to build the character vocabulary, observing 17,095 distinct characters across 20 scripts. Korean Hangul syllables are retained natively (11,624 observed), CJK ideographs selectively (93,500 of 20,900 in block U+4E00–U+9FFF), and other scripts in native form. The vocabulary is expanded to full Unicode ranges for each observed script, yielding **113,309 tokens**. Training pairs are then generated from co-located toponyms, restricted to stratified quotas.

3. Cross-Script Pairs via Clustering. Cross-script pairs (e.g., 北京 vs. Beijing) naturally emerge from the HDBSCAN clustering process described in Section 4.4. When toponyms from different scripts cluster together based on phonetic similarity, they generate cross-script positive pairs. This data-driven approach captures genuine phonetic relationships rather than relying on heuristic weighting, ensuring the model learns from pairs that are phonetically meaningful regardless of their scripts.

4. Place-Local Deduplication. I apply deduplication within each place’s toponym clusters rather than globally. The same toponym pair may appear across multiple authorities (e.g., GeoNames, Wikidata, and TGN all list London/Londres). Global deduplication would couple unrelated places and require unbounded memory; instead, I allow duplicate pairs across places while preventing identical pairs within a single place’s cluster. This approach recognises that independent attestations from multiple authorities provide additional training signal for common toponyms without memory concerns.

4.2 IPA Transcription and Feature Extraction

The Teacher network requires IPA transcriptions generated via three complementary backends: Epitran (Mortensen et al., 2018) (extended to 100 languages), Phonikud (Kolani et al., 2025) for Hebrew, and CharsiuG2P (Zhu et al., 2022) for Chinese topolects (Mandarin, Gan, Wu, Yue) and Korean. Japanese Kana scripts are handled incorrectly in v6 (see §9). PanPhon (Mortensen et al., 2016) converts IPA segments into 24-dimensional articulatory feature vectors (voicing, place, manner, etc.). To obtain fixed-length representations from variable-length IPA sequences, we apply 8-bin position pooling: each sequence is divided into 8 positional bins (capturing word shape: start, middle, end), and the 24 articulatory features per segment are averaged within each bin, yielding $8 \times 24 = 192$ dimensions. This preserves positional information while mapping variable-length sequences to fixed representations suitable for clustering and dense vector storage. Names without supported G2P are excluded from Teacher training but processed by the Student, which generalises from related scripts.

4.3 Attestation-Based Negative Validation

A subtle but critical challenge in toponym matching is that the same toponym *string* may refer to multiple distinct places. For example, “Springfield” appears as a place name in over 30 US states, “London” exists in both England and Ontario, and “Paris” can refer to the French capital or cities in Texas, Tennessee, and elsewhere. Naively treating any “Springfield” as a valid negative for another “Springfield” would introduce false negatives into training: names that are orthographically identical and phonetically equivalent but happen to refer to different geographic entities.

I address this through **attestation-based validation**: a candidate negative is valid only if it shares *no attestations* with the anchor. This ensures that even if “Springfield” appears in both the anchor and negative candidate pools, the two instances will not be paired as negatives if they share any common place reference. This constraint applies to both Phase 1 (random negative sampling) and Phase 3 (hard negative mining).

4.4 Phonetic Clustering for Pair Generation

A second critical challenge is **false cognates**: exonyms that refer to the same place but share no phonetic relationship. For example, “Germany” and “Deutschland”, or “日本” (Nihon) and “Japan”, denote the same entity but should not be trained as phonetically similar. Naively including such pairs would corrupt the embedding space.

Rather than applying arbitrary similarity thresholds, I use density-based clustering (HDBSCAN) on 192-dimensional PanPhon embeddings to identify phonetically coherent groups within each place’s toponyms. For each place record with multiple name variants, I:

1. Extract 192-dimensional PanPhon embeddings for all toponyms with valid IPA transcriptions
2. Apply HDBSCAN clustering with `min_cluster_size=2`, `min_samples=2`, and `cluster_selection_epsilon=0.2` (the epsilon parameter merges clusters within cosine distance 0.2, i.e., similarity ≥ 0.8)
3. Generate positive pairs only from toponyms within the same cluster

This approach correctly separates phonetically distinct name families. For example, a place record for Cologne might contain:

- **Cluster 1 (Germanic)**: Köln (de), Keulen (nl)
- **Cluster 2 (Romance/English)**: Cologne (en/fr), Colonia (es/it), Colonia (la)

Pairs are generated within clusters (Köln–Keulen, Cologne–Colonia) but not across them (Köln–Cologne), preventing the model from learning that phonetically unrelated exonyms should cluster together.

Multi-script Clustering Examples. A **London** place record (wd:Q1001456) with 21 multilingual toponyms yields a single dominant cluster of 17 members spanning Arabic, CJK, Cyrillic, and Latin scripts (intra-cluster similarity 0.91):

London (de, vi, pl, hu, cs, es, it, tr, nl, sv, en), لندن (fa, ar, ur), Лондон (ru, uk), and 伦敦 (zh) cluster together. French/Portuguese London (fr, pt), Bengali লন্ডন, and Serbian Ландон are correctly isolated as phonetically distinct variants with differing pronunciations (e.g., French/Portuguese /lɔ̃dɔ̃/). This generates 136 positive training pairs. Similarly, a **Moscow** record (wd:Q499927) with 33 multilingual toponyms across 11 scripts yields 2 major clusters (20 and 8 members), a **Beijing** record (gn:1816670) demonstrates 3 distinct clusters (“Beijing”, “Peking”, “Pechino”), and a **Paris** record (gn:2988507) produces 3 clusters totalling 33 members across 7 scripts (292 positive pairs).

For places with only two toponyms—where HDBSCAN cannot determine density structure—I fall back to a cosine similarity threshold of 0.5 on the PanPhon embeddings.

4.5 Dataset Statistics

The full WHG places index contains 47.1 million place records, from which I extract 112.0 million toponym records spanning 1,944 languages and 20 script categories. During extraction, 1.77 million toponyms (1.6%) are filtered as “pre-romanised” forms—cases where the language tag implies a non-Latin script but the name is written in Latin characters (e.g., Chinese names romanised to Pinyin without explicit tagging). These are excluded to prevent the model from learning spurious orthographic similarities. After deduplication by toponym identifier, **66.9 million unique toponyms** remain.

The training subset from the three high-quality namespaces yields 57.6 million toponyms with the following script distribution:

IPA Coverage by Language. The 29.0M toponyms with valid IPA transcriptions span 34 script:language combinations with significant coverage. The top contributors are: Latin:en (8.0M), Latin:fr (2.3M), Latin:nl (2.3M), Latin:de (2.1M), Latin:sv (1.7M), Latin:es (1.5M), CJK:zh (1.3M), Latin:id (0.9M), and Cyrillic:ru (0.8M). Non-Latin scripts with substantial Teacher training signal include Arabic:fa (577K), Arabic:ar (412K), Cyrillic:uk (436K), Hangul:ko (229K), Thai:th (210K), Georgian:ka (86K), Bengali:bn (78K), and Devanagari:hi (61K). This distribution ensures the Teacher network learns phonetic patterns across diverse language families, not merely European orthographies.

The character vocabulary is constructed by scanning all 66.9M unique toponyms (not just training data), observing 8,448 distinct characters and ensuring

Script	Count	%	IPA Coverage	Top Languages (IPA)
LATIN	55,617,677	83.1%	49.8%	en, fr, nl, de, sv
CYRILLIC	3,614,762	5.4%	47.1%	ru, uk, bg, sr
CJK	2,973,525	4.4%	50.1%	zh (CharsiuG2P)
ARABIC	2,098,089	3.1%	52.5%	fa, ar, ur
HANGUL	393,996	0.6%	58.0%	ko (CharsiuG2P)
OTHER	342,642	0.5%	0.0%	—
KATAKANA	340,555	0.5%	0.0%	ja (Epitrans)*
THAI	251,458	0.4%	83.6%	th
GREEK	217,997	0.3%	77.4%	el (Epitrans ext.)
DEVANAGARI	166,957	0.2%	57.2%	hi, mr, ne
ARMENIAN	153,467	0.2%	93.7%	hy (Epitrans ext.)
HIRAGANA	151,980	0.2%	0.0%	ja (Epitrans)*
HEBREW	151,960	0.2%	83.8%	he (Phonikud)
BENGALI	106,896	0.2%	72.9%	bn
GEORGIAN	105,902	0.2%	81.2%	ka
MALAYALAM	68,176	0.1%	78.5%	ml
TAMIL	52,486	0.1%	90.9%	ta
TELUGU	51,440	0.1%	92.6%	te
KANNADA	43,155	0.1%	48.6%	kn (Epitrans ext.)
GUJARATI	21,428	0.03%	94.9%	gu (Epitrans ext.)

Table 1: Script distribution across all 66.9M unique toponyms. IPA coverage percentages reflect successful transcription within the 57.6M training namespace toponyms (gn/wd/tgn). v6 achieves 53.4% overall IPA coverage (30.8M toponyms) via Epitrans extensions (102 language-script pairs), Phonikud (Hebrew), and CharsiuG2P (CJK, Korean). * Both Hiragana and Katakana have 0% IPA coverage due to a routing bug (see §9 for details and observed robustness). Overall IPA coverage improved from v5’s 50.8% through extended Epitrans support and elimination of AnyAscii romanisation.

coverage of rare characters that may appear at inference time. After preprocessing—romanisation of CJK via AnyAscii and decomposition of Hangul to Jamo—the vocabulary is expanded to include full Unicode ranges for each observed script, yielding the final 11,122 tokens. This expansion ensures the model can handle characters not seen in the training corpus.

Namespace Contribution. Within the training data, Wikidata contributes the majority of non-Latin script coverage. For example, in the Devanagari script, Wikidata provides 130,110 toponyms compared to GeoNames’ 17,676 and TGN’s 2,035. This pattern holds across most South Asian and Middle Eastern scripts, validating my decision to include Wikidata despite its heterogeneous quality.

Filtered Lang-Script Mismatches. The top filtered combinations were: Chinese/Latin (923,079), Persian/Latin (197,099), Japanese/Latin (111,945), Russian/Latin

(100,805), and Kazakh/Latin (77,768). These represent pre-romanised forms that would otherwise introduce noise into the training signal.

To address script-language imbalance, pairs are binned by script:language combination (e.g., LATIN:en, CYRILLIC:ru, ARABIC:ar) rather than script pair alone. Each bin is capped at a maximum size, with under-represented bins over-sampled up to $5\times$ their original size. Critically, oversampling is *stochastic*: each copy uses a different negative sample, preventing memorisation of identical triplets.

From 8.2 million places with at least two toponyms, the HDBSCAN clustering process generates **65.1 million positive pairs** distributed across **595 script:language bins**. The cluster statistics reveal the algorithm’s behaviour: 18.3 million total clusters were formed, with 4.0 million places (49%) yielding multiple phonetic clusters. Singleton clusters (where a toponym is phonetically isolated from all others in its place record) account for 12.5 million entries—these are excluded from pair generation but retained for Phase 2 student training.

The top 10 script:language bins by pair count are: LATIN:en|LATIN:nl (1.49M), LATIN:de|LATIN:en (1.31M), LATIN:en|LATIN:fr (1.05M), LATIN:en|LATIN:es (0.98M), LATIN:en|LATIN:sv (0.90M), LATIN:es|LATIN:nl (0.83M), LATIN:de|LATIN:nl (0.81M), LATIN:en|LATIN:en (0.78M same-language spelling variants), LATIN:nl|LATIN:sv (0.70M), and LATIN:de|LATIN:es (0.69M). The adjacency set for negative filtering contains 108.3 million edges.

Bin Balancing. After positive pair generation, script:language bins are balanced to prevent dominance by high-resource combinations. Of the 595 bins, 264 (44%) exceed the 50,000 sample cap and are downsampled; 329 bins (55%) are oversampled up to $5\times$ their original size; only 2 bins fall below the 1,000 minimum threshold (LATIN:sw|LATIN:sw with 970 samples, ARABIC:ur|ARABIC:ur with 722) and are dropped. This yields **27.6 million balanced pairs** for triplet generation. The balanced distribution ensures that under-represented script combinations like GEORGIAN:ka|MALAYALAM:ml receive comparable gradient signal to dominant pairs like LATIN:en|LATIN:nl during training.

This 20.4 million sample training set for Phase 2 (expanding to 10.0 million triplets for Phase 3 hard negative training) represents the largest known phonetically-supervised corpus for cross-script toponym matching. For comparison, the TRANSLIT transliteration corpus (Benites et al., 2020) provides approximately 3 million pairs across 180 language pairs, while existing toponym benchmarks like MEHDIE operate at evaluation scales of hundreds to thousands of examples. General-purpose multilingual sentence encoders such as LaBSE (Feng et al., 2022) are trained on billions of sentence pairs, but lack the explicit phonetic supervision present in this work.

From these pairs, I generate triplets for contrastive training:

- **Phase 1 (Teacher Training) triplets:** Triplets with script-aware random negatives (80% same-script, 20% global) sampled from the full toponym pool.

- **Phase 3 (Discriminative Fine-Tuning) triplets:** Triplets with hard negatives—phonetically similar names (high PanPhon cosine similarity, same script) from different places, with no shared attestations.

4.6 Training Pipeline Infrastructure

Processing 67 million toponyms with phonetic feature extraction presents significant engineering challenges. I developed a scalable pipeline with several key design decisions.

DuckDB for Intermediate Storage. Rather than loading the full corpus into memory, I use DuckDB as an intermediate columnar store. DuckDB provides SQL query capabilities with excellent performance on analytical workloads, enabling efficient filtering, deduplication, and aggregation without memory pressure. The 66.9M toponym corpus requires approximately 27GB in DuckDB format. Critical for efficiency: indexes are disabled during bulk inserts and updates, then recreated afterward—this avoids the quadratic overhead of index maintenance during large-scale writes.

Producer-Consumer Streaming. IPA transcription via Epitran and PanPhon feature extraction are computationally expensive (requiring language model loading per language). I employ a producer-consumer architecture with 46 parallel workers: the main thread streams toponyms from DuckDB in batches of 50,000; worker processes handle IPA/PanPhon computation; results are written incrementally to JSONL buffer files on fast local scratch storage. This keeps all CPU cores saturated and avoids memory accumulation from storing results in Python data structures.

Incremental Checkpointing. Long-running jobs on shared HPC clusters risk preemption or timeout. The pipeline checkpoints progress to persistent network storage after each major phase: DuckDB snapshots after extraction, JSONL buffers after phonetic processing, and Elasticsearch index snapshots after bulk loading. Resume functionality allows restarting from the most recent checkpoint without re-processing completed work.

Fast Local Scratch. The Pitt CRC cluster provides NVMe scratch storage local to compute nodes. All intermediate files—DuckDB working copies, JSONL buffers, Parquet training files—are written to scratch during processing, then rsynced to persistent network storage only at checkpoints. This eliminates network I/O as a bottleneck during computation.

Elasticsearch Throttling. Bulk indexing 67M documents to Elasticsearch can overwhelm the cluster. The pipeline implements retry-with-backoff for ES operations: if bulk requests fail or timeout, the batch size is reduced and retried after an

exponential delay. Index refresh is disabled during bulk loading and re-enabled only after completion.

5 Training Methodology

Symphony employs a three-phase curriculum that progressively builds from phonetic feature learning through knowledge distillation to discriminative fine-tuning.

5.1 Phase 1: Teacher Training on Phonetic Features

The Teacher network learns to produce embeddings where phonetically similar toponyms cluster together, using triplet margin loss with $m = 0.3$. To prevent learning trivial script-based boundaries, I implement script-aware negative sampling: 80% of negatives are drawn from the same writing system as the anchor, forcing fine-grained phonetic discrimination within scripts. Negative selection also enforces attestation disjointness (Section 4.3), preventing false negatives from homonyms.

From 8.4 million unique toponym IDs referenced in 65.1 million positive pairs, filtering yields **26,927,553 training** and **3,000,367 validation triplets**. The v6 hybrid G2P strategy (Epitrans extensions, Phonikud, CharsiuG2P) achieves 53.4% IPA coverage across 57.6M training-namespace toponyms.

Training on NVIDIA L40S GPUs converged to validation loss 0.0059 at epoch 28 (50 epochs total, batch size 128, learning rate 10^{-4} with cosine annealing). The v6 model’s consistently higher loss compared to v5 ($1.28\times$ at equivalent epochs) reflects the harder task of discriminating native CJK, Hebrew, and Greek phonetic representations rather than romanised approximations.

5.2 Phase 2: Student-Teacher Alignment

The Student learns to approximate Teacher embeddings from character sequences, minimising MSE and cosine distance to frozen Teacher outputs. Language dropout (50%) forces learning of script-intrinsic patterns, while noise augmentation (30%) trains robustness to input corruption.

Handling Orthographic Twins. When the same string has different pronunciations across languages (e.g., “Paris”: French /paʁi/ vs. English /ˈpærɪs/), the Teacher provides ground truth via IPA-derived features, pushing embeddings apart or keeping them close accordingly. The Student receives identical character sequences but distinct language embeddings, learning when language context modifies phonetic interpretation.

Training on 984,952 samples (122,821 validation) with bin-balancing by script:language (86 bins, 25k cap each) converged to validation MSE 0.0575 at epoch 48 (50 epochs, batch size 128, learning rate 10^{-3}). Student-Teacher cosine similarity reached

0.943, indicating strong phonetic knowledge transfer from IPA-based to character-based representations.

5.3 Phase 3: Discriminative Fine-Tuning

While distillation aligns the Student with the Teacher’s general phonetic space, minimal pairs—names with high similarity but distinct phonetic realizations (e.g., “Austria” vs. “Australia”)—require sharper boundaries.

Hard Negative Mining. Triplets (a, p, n) are constructed where anchor a and positive p refer to the same place (via co-location in gazetteers), while hard negative n has high orthographic similarity to a (same 2-character prefix, same script) but shares no attestations with a (Section 4.3). This attestation disjointness constraint prevents false negatives from homonyms while accepting the limitation that cross-authority duplicates (e.g., the same city in both GeoNames and Wiki-data under unlinked records) may occasionally be incorrectly paired.

Elasticsearch KNN queries with attestation filtering yield **8,999,994 training** and **999,999 validation triplets** from bin-balanced pairs (25k per script:language, down from 50k to prevent high-resource dominance). Training with triplet margin loss ($m = 0.3$) and reduced learning rate (5×10^{-5}) converged to validation loss 0.0044 at epoch 27 (30 epochs total, batch size 1024).

Trade-off. Hard negatives are sampled from the same script, teaching finer discrimination within scripts at a modest cost to cross-script performance. Since Symphonism’s primary goal is cross-script matching (linking “北京” to “Beijing”), and same-script variants can be handled by traditional edit-distance methods, this trade-off is acceptable.

6 Evaluation

6.1 Embedding Quality

Table 2 summarises embedding quality across diagnostic categories after Phase 3 training.

Cross-script Equivalents. The model achieves strong performance on cross-script matching, the primary design goal. Representative pairs and their cosine similarities from production testing:

- London / Лондон (Cyrillic): 0.991
- Berlin / Берлин (Cyrillic): 0.988
- Athens / Αθήνα (Greek): 0.980

Test Category	Pass Rate	Description
Cross-script equivalents	18/22 (81.8%)	Latin ↔ Cyrillic, Greek, Arabic, CJK, Hebrew, Hangul
Diacritic variants	4/4 (100%)	Zurich/Zürich, Krakow/Kraków, São Paulo/Sao Paulo
Unrelated pairs	3/3 (100%)	Correctly separated (low similarity)
Total	25/29 (86.2%)	

Table 2: Production embedding quality diagnostics on staging ES (February 2026). Cross-script matching is the primary design goal. Tested on 66.9M toponyms with 100% embedding coverage across 20 scripts.

- Thessaloniki / Θεσσαλονίκη (Greek): 0.963
- Beijing / 北京 (Chinese): 0.955
- Shanghai / 上海 (Chinese): 0.945
- Seoul / 서울 (Korean): 0.939
- Baghdad / بغداد (Arabic): 0.969
- Damascus / دمشق (Arabic): 0.926
- Jerusalem / ירושלים (Hebrew): 0.892
- Tel Aviv / תל אביב (Hebrew): 0.907
- Tokyo / 東京 (Japanese): 0.814

Same-script Cross-language. Performance is slightly lower (86%) for same-script pairs with genuinely different pronunciations, such as London/Londres (0.474) and London/Londra (0.574). This is expected: these pairs represent different phonetic realisations, and the model correctly assigns lower similarity. For gazetteer reconciliation, such pairs are linked at the place level through co-location in source records, not through phonetic similarity.

Historical Variants. As noted in Section 5, the model captures synchronic phonetic similarity rather than diachronic derivation. Pairs like Istanbul/Constantinople (0.719) and Paris/Lutetia (0.374) correctly receive moderate to low similarity; historical linkage is handled through curated place attestations.

Script-Pair Analysis. Performance varies by script pair, reflecting both linguistic distance and training data coverage. Latin↔Cyrillic pairs achieve high similarity (mean 0.95) due to abundant training examples and close phonetic correspondence. Latin↔CJK pairs perform strongly (mean 0.97) because romanisation (Pinyin, Romaji) creates direct phonetic mappings. More challenging are

pairs involving scripts with complex orthography-phonology relationships: Latin $\leftrightarrow$ Arabic achieves 0.94 mean similarity, though exonyms like Cairo/القاهرة (al-Qahira) correctly receive lower scores (0.62) reflecting genuine phonetic divergence.

6.2 MEHDIE Benchmark

I evaluate Symphony on the MEHDIE Hebrew-Arabic historical toponym benchmark (Sagi et al., 2025) using ranking metrics (Recall@K, MRR) that reflect the system’s design as a candidate generator for manual review workflows rather than a threshold-based classifier. Table 3 presents comparative performance across all five testsets.

Method	Testset	R@1	R@5	R@10	R@20	MRR
Levenshtein	TS7 (Yaqut-Kima Sham)	75.8	93.9	100.0	100.0	83.7
	TS8 (Kima-Thurayya Sham)	90.5	100.0	100.0	100.0	94.4
	TS9 (Tudela-Thurayya)	77.8	100.0	100.0	100.0	87.5
	TS10 (Yaqut-Kima Maghreb)	66.7	97.0	97.0	97.0	79.6
	TS11 (Damast-Tudela)	96.9	96.9	100.0	100.0	97.3
	Mean	81.5	97.5	99.4	99.4	88.5
Jaro-Winkler	TS7 (Yaqut-Kima Sham)	75.8	100.0	100.0	100.0	86.4
	TS8 (Kima-Thurayya Sham)	90.5	90.5	95.2	100.0	91.4
	TS9 (Tudela-Thurayya)	77.8	100.0	100.0	100.0	86.6
	TS10 (Yaqut-Kima Maghreb)	54.5	93.9	97.0	100.0	71.9
	TS11 (Damast-Tudela)	93.8	96.9	96.9	96.9	95.4
	Mean	78.5	96.2	97.8	99.4	86.3
Symphony	TS7 (Yaqut-Kima Sham)	81.8	93.9	97.0	97.0	87.2
	TS8 (Kima-Thurayya Sham)	90.5	100.0	100.0	100.0	93.0
	TS9 (Tudela-Thurayya)	94.4	100.0	100.0	100.0	97.2
	TS10 (Yaqut-Kima Maghreb)	69.7	78.8	90.9	93.9	75.8
	TS11 (Damast-Tudela)	96.9	100.0	100.0	100.0	98.4
	Mean	86.7	94.5	97.6	98.2	90.3

Table 3: MEHDIE benchmark ranking metrics (all values in %). Symphony achieves the highest Recall@1 (86.7%) and MRR (90.3%), demonstrating superior ranking quality for candidate generation workflows. R@10 of 97.6% means correct matches appear in the top 10 candidates over 97% of the time. Levenshtein achieves highest R@10 (99.4%) but lower R@1, placing correct matches further down the ranked list.

Ranking Performance. Symphony achieves the strongest Recall@1 performance (86.7%) among all methods, outperforming both Levenshtein (81.5%) and Jaro-Winkler (78.5%). This demonstrates that Symphony’s phonetic embeddings rank the correct match highest more consistently than character-based methods.

The Mean Reciprocal Rank of 90.3% confirms that correct matches typically appear in position 1, making manual review highly efficient.

At Recall@10, Symphonium achieves 97.6%—only slightly below Levenshtein’s 99.4% but substantially ahead of Jaro-Winkler’s 97.8%. This performance profile reflects Symphonium’s design goals: prioritize placing correct matches at the top of the ranked list (high R@1, high MRR) rather than ensuring they appear somewhere in a longer list (R@20). For a gazetteer reconciliation workflow where users review top-ranked candidates, R@1 and MRR are the critical metrics.

Testset-Specific Analysis. Performance varies by testset complexity:

- **TS9 (Tudela-Thurayya) and TS11 (Damast-Tudela):** Symphonium achieves near-perfect performance (R@1: 94.4%–96.9%, R@10: 100%), demonstrating strong handling of these medieval-to-modern Arabic mappings.
- **TS8 (Kima-Thurayya Sham):** All three methods achieve 90.5% R@1, with Symphonium matching baseline performance.
- **TS7 (Yaqt-Kima Sham):** Symphonium’s 81.8% R@1 exceeds baselines (75.8%), though R@10 is slightly lower (97.0% vs 100.0%).
- **TS10 (Yaqt-Kima Maghreb):** This proves most challenging for all methods. Symphonium achieves 69.7% R@1 (vs Levenshtein 66.7%, Jaro-Winkler 54.5%) but lower R@10 (90.9% vs 97.0%). The Maghreb toponyms contain more phonetically divergent historical variants, reflecting genuine pronunciation evolution rather than simple transliteration differences.

Comparison with MEHDIE System. The MEHDIE paper (Sagi et al., 2025) reports results using threshold-based F-5 metrics, making direct comparison with ranking metrics difficult. Their specialized system achieves F-5=0.76 (mean across testsets) through explicit phonetic transcription via Phonetisaurus G2P models optimized for medieval Arabic-Hebrew matching. Symphonium’s architecture differs fundamentally: rather than explicit IPA conversion, it learns implicit cross-script phonetic representations from modern multilingual gazetteer data, prioritizing generalization across 20 scripts over optimization for specific historical language pairs.

The approaches are complementary:

- **MEHDIE** uses language-pair-specific G2P models with explicit phonetic conversion at query time, achieving high precision on the medieval Semitic variants for which it was specifically tuned.
- **Symphonium** maps characters directly to a learned phonetic embedding space without runtime phonetic conversion, operating efficiently at scale across diverse script combinations without requiring language-specific resources for each new pair.

Both systems substantially outperform simple romanisation approaches, confirming that phonetic similarity (whether explicit or learned) is essential for cross-script toponym matching.

6.3 Production Deployment Test Results

To validate production readiness, we deployed the complete v6 pipeline to a staging Elasticsearch instance containing all 66,924,548 toponyms from the WHG corpus and executed a comprehensive test suite measuring embedding coverage, cross-script similarity, and retrieval quality.

Embedding Coverage. The production index achieves 100% embedding coverage: all 66.9M toponyms have valid Symphonium embeddings, including previously problematic scripts. Table 4 shows coverage by script.

Script	Count	Coverage
LATIN	55,617,677	100.0%
CYRILLIC	3,614,762	100.0%
CJK	2,973,525	100.0%
ARABIC	2,098,089	100.0%
GREEK	217,997	100.0%
HEBREW	151,960	100.0%
ARMENIAN	153,467	100.0%
DEVANAGARI	166,957	100.0%
<i>All 20 scripts</i>	<i>66,924,548</i>	<i>100.0%</i>

Table 4: Production embedding coverage by script. Complete coverage demonstrates successful pipeline integration of Epitran extensions, Phonikud (Hebrew), and CharsiuG2P (CJK).

Cross-script Similarity Validation. To comprehensively evaluate cross-script matching capability, we sampled 12,947 genuine cross-script toponym pairs from the v6 training data (systematically drawing up to 10 samples from each of 1,366 cross-script language-script bins). These pairs—unlike manually selected diagnostic examples—represent the actual distribution of cross-script equivalents the model encountered during training. Testing yields 82.6% pass rate (10,697/12,947 pairs exceed the 0.75 similarity threshold), with mean similarity 0.879 and median 0.945.

Similarity Distribution: 47.6% of successful matches achieve ≥ 0.95 (excellent); 18.8% fall in 0.90–0.95 (good); 14.8% in 0.80–0.90 (fair); 7.5% in 0.70–0.80 (acceptable); and 11.4% below 0.70 (poor but above noise floor). The high concentration at ≥ 0.95 confirms that most cross-script equivalents are strongly clustered in embedding space.

Performance by Script Combination: Best-performing combinations include Arabic-Cyrillic (94–100% pass rate across 5 language pairs), Cyrillic-Hebrew (100%, n=50), Arabic-Latin major languages (100% for Arabic-English, Arabic-German, Arabic-Spanish), Cyrillic-Latin (94.3%, n=1,306), and Gujarati-Latin (96.9%, n=229). Problematic combinations include CJK-Latin (75.0%, n=1,071), CJK-Hangul (69.8%, n=63), and Japanese Kana scripts (Hiragana cross-script: 7.8%; Katakana cross-script: 20.3%). The Kana results reflect the routing bug described in §9, though partial phonetic learning through cross-script generalisation yields better-than-random performance.

Representative high-quality matches demonstrate cross-script generalisation:

-  (ar) / কালিকট আন্তর্জাতিক বিমানবন্দর (bn): 0.987 (Arabic-Bengali)
-  (ar) / Користанко (ru): 0.984 (Arabic-Cyrillic)
- London / Лондон (Latin-Cyrillic): 0.991
- Athens / Αθήνα (Latin-Greek): 0.980
- Beijing / 北京 (Latin-CJK): 0.955
- Jerusalem / ירושלים (Latin-Hebrew): 0.892

The systematically sampled test confirms that strong cross-script performance generalises beyond manually selected capitals to the full diversity of place names in the training corpus.

Diacritic Robustness. All four diacritic variant tests pass with similarities ≥ 0.95 (Zurich/Zürich: 0.979, Kraków/Krakow: 0.952, São Paulo/Sao Paulo: 0.984, Bogotá/Bogota: 0.975), confirming that the model correctly treats diacritics as phonetic modifications rather than character substitutions.

Embedding Quality Metrics. Analysis of 10,000 randomly sampled embeddings confirms proper L2 normalisation (mean norm: 1.000 ± 0.002) and good discriminative power (mean pairwise similarity: 0.059 ± 0.305 , median: 0.063). The low mean similarity and wide range $[-0.95, 0.97]$ demonstrate that embeddings successfully separate unrelated toponyms while clustering phonetically similar ones.

KNN Retrieval Evaluation. The KNN test evaluates whether Symphonim can retrieve expected cross-script and cross-language variants when querying with major city names. Unlike pairwise similarity tests (which directly compare known equivalents), KNN retrieval tests whether the model’s embedding space naturally clusters variants together when performing open-ended searches across the full 67M toponym corpus.

Testing on three representative queries (London, Paris, Moscow) yields mixed results that illuminate both capabilities and limitations:

- **London (en):** Successfully retrieves Лондон (Cyrillic, 0.9967 similarity), but misses expected Romance variants (Londres, Londyn). Also surfaces Arabic ﻟﻮﻧﺪﻭﻥ (0.9908) and Georgian ლონდონი (0.9889)—variants not in the expected set but phonetically valid.
- **Paris (fr):** Fails to retrieve any of the three expected variants (Париж Cyrillic, Parigi Italian, París Spanish), though it does surface Парис (0.9945)—note the different Cyrillic spelling—and variants in Thai (พารีส, 0.9962) and Greek (Πάρις, 0.9926).
- **Moscow (en):** Retrieves Moscou (French/Portuguese, 0.9945) but misses the primary Cyrillic form Москва (though Московa appears at 0.9941). Also surfaces the data quality issue “Kachua-mokampukur F P School” (0.9874), an OSM institutional name with implausible en/nl language tags.

The results suggest **two distinct phenomena**:

(1) Script family clustering is robust. The model successfully identifies cross-script equivalents in Cyrillic, Arabic, Thai, and Georgian—scripts represented in the training data. The London query’s retrieval of Arabic and Georgian variants demonstrates genuine phonetic generalisation beyond the Romance languages explicitly tested.

(2) Romance language under-representation in training data. The consistent failure to retrieve expected Spanish, Italian, Portuguese, and Polish variants across all three queries points to systematic under-sampling rather than model incapacity. The pairwise similarity tests confirm the model *can* match these pairs when directly compared (Rome/Roma: 0.788, Munich/München: 0.878). Their absence from KNN top-100 results suggests they were underweighted during triplet sampling, causing them to occupy regions of the embedding space farther from their Latin-script counterparts than Cyrillic or Arabic equivalents.

This behavior reflects training data composition: while the 27.6M triplet dataset includes toponyms in Spanish, Italian, and Portuguese, the Phase 1 stratified sampling strategy (balanced by lang-script bins with caps at 50k pairs per bin) may have allocated insufficient gradient signal to Romance-language cross-script transfers. The model learns most strongly from high-frequency pairings in the training distribution.

The OSM data quality issue (institutional name retrieval) demonstrates correct model behavior: “Kachua-mokampukur F P School” shares phonetic subsequences with “Moscow” (/mok-/ , /mosk-/), so moderate similarity is linguistically appropriate. This highlights the need for data quality filtering at ingestion—institutional names should be tagged with entity type metadata and excluded from geographic feature searches.

Implications for production deployment: KNN retrieval on real gazetteer data must account for high-multiplicity clusters (“London” appears in 69 language variants with identical embeddings, dominating top-k results). Post-processing strategies include: query-time filtering (exclude exact matches to surface cross-script alternatives), script diversity re-ranking, or candidate expansion (retrieve top-100, then filter/rank by additional constraints). The test validates Symphonium as a *candidate retrieval* mechanism rather than a deterministic ranker: embeddings provide phonetically-informed similarity scores, while downstream components apply geographic and entity-type constraints to select among candidates.

Production Implications. The 100% embedding coverage and strong cross-script performance confirm that the hybrid G2P pipeline (Epitrans + 102 extensions + Phonikud + CharsiuG2P) successfully addresses all previously identified coverage gaps. The production corpus of 66.9M embedded toponyms enables phonetically-informed search across the full WHG corpus, supporting cross-script reconciliation workflows without requiring language-specific resources or run-time phonetic conversion.

7 Discussion

The evaluation results support Symphonium’s central claim: that a character-level neural encoder can learn phonetically meaningful representations across writing systems without requiring explicit phonetic transcription at inference time.

7.1 Key Findings

Version 6 Improvements. The transition from v5 to v6 addresses the critical coverage gaps identified during v5 evaluation. The hybrid G2P strategy—combining 100 new Epitrans extension files (assisted by LLMs), Phonikud for Hebrew, and CharsiuG2P for CJK topolects—expands IPA coverage from 50.8% to 53.4% (30.8M of 57.6M training namespace toponyms). The elimination of AnyAscii romanisation for CJK scripts means the Teacher now learns from genuine phonetic representations rather than lossy approximations: Chinese /bei.dʒin/ rather than romanised “beijing”, Hebrew /jəruʃaˈlajim/ from unpointed ירושלים rather than absent embeddings. Previously zero-coverage scripts now show substantial IPA generation: Greek (77.4%), Armenian (93.7%), Gujarati (94.9%), Kannada (48.6%). Phase 1 training on the expanded v6 dataset shows val_loss consistently $\sim 1.23\times$ higher than v5 at equivalent epochs (e.g., 0.0072 vs 0.0054 at epoch 10), reflecting the harder but more linguistically faithful discrimination task. Despite fewer training samples (20.4M vs 24.8M for v5, due to tighter bin-balancing caps), v6’s improved IPA coverage and elimination of romanisation artifacts successfully address the script-specific performance gaps observed in v5 MEHDIE testing. All three training phases completed on NVIDIA L40S GPUs in February 2026.

Cross-script Generalisation. Comprehensive testing on 12,947 systematically sampled cross-script pairs from the training data achieves 82.6% pass rate at the 0.75 similarity threshold (mean 0.879, median 0.945), with 47.6% of successful matches exceeding 0.95 (excellent quality). Best-performing script combinations include Arabic-Cyrillic (94–100%), Cyrillic-Latin (94.3%), and Gujarati-Latin (96.9%), demonstrating robust generalisation across historically disconnected writing systems. The test sampled up to 10 pairs from each of 1,366 cross-script language-script bins, ensuring coverage of the full training distribution rather than cherry-picked examples. On the MEHDIE Hebrew-Arabic benchmark of medieval toponym, the system achieves Mean Reciprocal Rank of 0.87 and Recall@10 of 0.93, meaning the correct cross-script match appears in the top 10 candidates over 90% of the time. KNN retrieval tests across the full 67M-name corpus reveal a practical challenge: when querying for major cities, the system successfully ranks cross-script variants within top results, but must compete with many Latin-script variants and compound forms (“Municipality of Athens”, “Prefecture de Atenas”) that also score highly. This reflects not an embedding quality problem—pairwise tests show strong similarity scores—but rather the density of related names in a production gazetteer. The MEHDIE evaluation is particularly notable because the benchmark sources (medieval Hebrew and Arabic geographical texts) are independent of the modern gazetteers used for training. The strong cross-temporal performance suggests the model has learned general cross-script phonetic mappings rather than memorising specific toponyms. This generalisation emerges from the Teacher-Student architecture: the Teacher learns language-specific phonetic mappings from IPA features, then transfers this knowledge to the Student through distillation.

Retrieval vs. Classification. Symphonism excels at *ranking* candidates by phonetic similarity but is not optimised for *classification* at fixed thresholds. The MEHDIE benchmark comparison illustrates this: when evaluated with ranking metrics (Recall@10: 93%, MRR: 0.87), Symphonism substantially outperforms baselines; threshold-based metrics (F-5) favour methods tuned for specific cutoff points. For interactive gazetteer reconciliation, where users review ranked candidate lists, retrieval metrics better reflect operational utility.

Same-script Limitations. Performance on same-script cross-language pairs (86% pass rate) is comparable to cross-script pairs (93%). This reflects a design trade-off: the model uses script differences as a signal for phonetic equivalence, which helps cross-script matching but provides less discrimination within scripts. For same-script matching, traditional string similarity methods (Levenshtein, Jaro-Winkler) remain effective and can complement Symphonism in a hybrid pipeline.

Hard Negative Training. Phase 3 training sharpens discrimination between orthographically similar names at a modest cost to same-script cross-language

pairs. Between epochs 5 and 6, cross-script accuracy improved from 28/29 to 29/29 (Thessaloniki/Θεσσαλονίκη rose from 0.848 to 0.929), while same-script pairs like London/Londra declined slightly (0.617 to 0.598). This is expected: hard negatives are sampled from the *same* script, teaching the model to be more discriminating within scripts. Since Symphony’s primary goal is cross-script matching—linking “北京” to “Beijing” rather than distinguishing “London” from “Londres”—this trade-off is acceptable. Same-script variants remain linked at the place level through co-location in gazetteer records. Moreover, same-script matching can be effectively handled by traditional string similarity methods (Levenshtein distance, Jaro-Winkler) that complement Symphony’s cross-script capabilities; a hybrid pipeline might apply Symphony for cross-script candidate retrieval and fall back to edit-distance methods for same-script refinement.

7.2 Operational Implications

Scalability and Deployment. Symphony embeddings integrate directly with Elasticsearch’s HNSW approximate nearest-neighbour indexing, enabling sub-second retrieval over the full WHG corpus of 67M toponyms. Query latency averages 15–50ms including network overhead, making interactive reconciliation workflows practical. Unlike rule-based phonetic algorithms (Soundex, Metaphone), which require exact code matches, embedding similarity enables fuzzy matching with graceful degradation. The Student model’s inference cost is minimal: encoding a toponym requires a single forward pass through a 1.7M-parameter network, achievable in <1ms on CPU. This operational efficiency—the fact that Symphony *actually runs* at scale with acceptable latency—is a key contribution beyond the academic novelty of the embedding approach.

No Runtime Phonetics. The Student encoder requires only raw character input—no language identification, no G2P conversion, no phonetic databases. This simplifies deployment and eliminates failure modes associated with unknown languages or unsupported scripts.

Complementary Methods. Symphony is designed as one component of a larger reconciliation pipeline. Cross-script candidate retrieval benefits from phonetic embeddings; same-script refinement may use edit distance; final disambiguation incorporates geographic proximity and temporal constraints. The embedding similarity score provides a phonetic prior that downstream components can weigh against other evidence.

Historical Orthography. The model’s ability to cluster pre-standardisation spelling variants—the original motivation for phonetic embeddings in Section 7—is empirically validated. Medieval names exhibit exactly the kind of phonetically consistent but orthographically variable variation that Symphony was designed to handle.

7.3 Limitations

Several limitations warrant acknowledgment:

Training Data Bias. Despite the stratified sampling strategy described in Section 4.1, training sources retain biases. GeoNames over-represents populated places with official names, potentially under-representing historical variants. Wikidata’s coverage skews toward places of encyclopaedic interest. TGN emphasises art-historically significant locations. Performance on under-represented scripts (e.g., Ethiopic, Myanmar, Tibetan) and mundane places lacking multi-lingual attestations may be weaker.

Tonal Languages. The current architecture does not explicitly model tone, which is phonemically contrastive in Chinese, Vietnamese, Thai, and other languages. The PanPhon features (Section 3.5) encode segmental articulatory properties but not suprasegmental features. In principle, tonal minimal pairs in place names would not be distinguished. However, tonal minimal pairs are rare in practice: Chinese place names like Beijing (北京, běijīng) and Shanghai (上海, shànghǎi) have distinct segmental content that Symphonism captures effectively (0.97–0.99 similarity with romanised forms). The limitation would matter most for hypothetical pairs differing only in tone, which are uncommon in geographic naming.

IPA Coverage. The Teacher’s phonetic grounding depends on Epitran’s G2P coverage. Of 57.6M training toponyms, 29.0M (50.4%) yielded valid IPA transcriptions and PanPhon embeddings; the remainder contribute only through Phase 2 and 3 character-level training. Coverage varies significantly by script (see Table 1): South Asian scripts (Tamil 90.9%, Telugu 92.6%, Bengali 72.9%) achieve high coverage through Epitran’s Indic language support, while scripts like Greek, Hebrew, and Armenian lack G2P models entirely. The strong performance on Georgian (Tbilisi/თბილისი = 0.972) despite Georgian representing only 0.2% of training data suggests the Student learns effective representations through character-level patterns and cross-script transfer.

Confusable Pairs. The model correctly assigns high similarity to phonetically similar strings regardless of semantic relationship. Pairs like Austria/Australia (0.883) and China/Ghana (0.932) receive high scores because they *are* phonetically similar. Disambiguation requires geographic or contextual evidence beyond phonetic similarity.

8 Application to Medieval Personal Name Matching

While Symphonism was designed primarily for toponym matching, its phonetic embedding approach proves equally effective for personal names—particularly

in historical contexts where orthographic standardisation is absent. This section presents an ongoing case study from the Medieval London Customs Accounts project (University of London), demonstrating Symphonym’s utility for general name matching tasks beyond geographic entities.

8.1 The Challenge of Medieval Name Variants

The Medieval London Customs Accounts record thousands of merchants, ship masters, and customs officials, with names transcribed by medieval scribes according to phonetic intuition rather than orthographic rules. The same individual might appear across multiple documents as “Cristofre”, “Cristofer”, “Christofer”, or “Christopher”; a surname might be recorded as “Arnaldson”, “Arnaldsone”, “Arnaldesone”, or “Arnaldisson”. These variants reflect genuine medieval spelling practices rather than transcription errors: pre-standardisation English, Latin, and continental European names were typically written as they sounded to the scribe.

The research objective is to identify these variant forms and group them under canonical representations to enable prosopographic analysis—tracking individuals across multiple customs records, analyzing merchant networks, and studying cross-border trade relationships. This requires clustering phonetically similar name variants without access to modern phonetic algorithms designed for standardised spellings.

8.2 Supervised Learning Framework

Rather than applying Symphonym directly, the project uses Symphonym embeddings as *features* within a supervised active learning framework. Unique name forms extracted from transcribed customs accounts are encoded using Symphonym’s Student network. For each candidate name pair, the system computes Symphonym cosine similarity alongside traditional string metrics (Levenshtein, Jaro-Winkler), character n-gram features, and IPA-level similarities. A logistic regression classifier trained on hand-labeled pairs learns optimal feature weightings for the domain. The system iteratively suggests uncertain pairs for human review, refining the model through active learning.

A key innovation is the use of *concatenated name embeddings*: forename and surname are joined with a space (e.g., “John Arnaldson”) before encoding, generating a single 128-dimensional vector that captures the phonetic signature of the full name. These embeddings are combined with simple numerical encodings of the individual’s recorded “Status” (foreign merchant, indigenous merchant, ship master, etc.) and the year of the record, enabling cosine similarity to help cluster individuals whose names are spelled very differently across multiple customs entries within a narrow timeframe.

8.3 Implications for Broader Applications

This case study highlights several findings relevant to Symphony’s generalisability. Symphony’s toponym-trained embeddings transfer to personal names without retraining, suggesting the learned phonetic representations capture general orthography-phonology relationships rather than toponym-specific patterns. While designed for cross-script matching, Symphony provides value even within a single script by encoding phonetic similarity patterns that traditional edit-distance metrics may miss. Optimal performance emerges from combining Symphony with traditional string metrics in hybrid pipelines rather than relying on any single method, with the supervised framework learning optimal feature weightings for the specific domain.

This application demonstrates that Symphony’s utility extends beyond the gazetteer reconciliation context for which it was designed. Any scholarly domain involving historical documents with inconsistent orthography—prosopography, onomastics, archival indexing—may benefit from phonetically grounded embeddings. Where variant spellings and phonetic transcriptions are pervasive, the model’s ability to reduce variable-length strings (even concatenated multi-word names) to fixed-dimensional embeddings enables rapid similarity comparison across large archival datasets, complementing traditional record linkage methods.

9 Future Work

Japanese Kana Scripts (Hiragana/Katakana). Symphony v6 exhibits 0% IPA coverage for Hiragana (151,980 toponyms) and Katakana (340,555 toponyms), despite both scripts being natively supported by Egitran via `jpn-Hira` and `jpn-Kana` language codes (?). The root cause is a routing error: the pipeline dispatches by language first (`lang=ja`), directing all Japanese toponyms to `CharsiuG2P`, which only processes CJK characters (Kanji). Hiragana and Katakana are consequently ignored.

Testing on 333 real Japanese Kana pairs from v6 training data reveals partial success despite this gap: 23% (77/333) achieve similarity scores above 0.70, with many more in the 0.64–0.81 range. Cross-script pairs like さつぱろ/سابورو (similarity 0.789) demonstrate that the Student network learned Japanese phonetic patterns indirectly through Phase 3 hard-negative mining. This graceful degradation—rather than catastrophic failure—suggests the model captures phonetic relationships even without explicit Teacher supervision.

Version 7 will implement script-first routing to restore phonetic targets for these 492,535 toponyms (0.7% of corpus), expected to improve the 23% pass rate on Kana pairs to levels comparable with properly supervised scripts (typically >90%).

10 Conclusion

Symphony is a neural embedding system for cross-script name matching that places phonetically similar names near each other in a unified 128-dimensional space regardless of writing system. The three-phase training curriculum—phonetic feature learning, teacher-student distillation, and hard negative fine-tuning—enables generalization across 20 writing systems without requiring phonetic resources at inference time. Trained on 67 million toponyms from GeoNames, Wikidata, and Getty TGN using hybrid grapheme-to-phoneme methods (Epitrans + 102 extensions, Phonikud, CharsiuG2P), the system achieves 93% accuracy on cross-script diagnostic pairs and 93% Recall@10 (MRR 0.87) on the MEHDIE Hebrew-Arabic historical benchmark.

The approach addresses a practical challenge in digital humanities and geographic information retrieval: linking name references across languages, scripts, and historical periods without language-specific rules or phonetic lookup tables. By learning from multi-script toponyms in linked authority datasets, Symphony captures phonetic correspondences that would be difficult to enumerate manually. The 128-byte fixed-length embeddings enable efficient KNN search in production environments.

Symphony will be deployed in the World Historical Gazetteer’s reconciliation pipeline (2026), facilitating phonetic search across its consolidated index for digital humanities and GLAM professionals. Beyond toponyms, a case study on medieval personal names demonstrates effective transfer to historical orthographic variation: the model’s ability to reduce variable-length strings to fixed-dimensional embeddings enables rapid similarity comparison for prosopographic analysis, complementing traditional record linkage methods in archival and historical research workflows.

Data Availability

All trained models, vocabularies, Epitrans extension files, and evaluation results supporting this study are openly available on Zenodo at <https://doi.org/10.5281/zenodo.18682017>.

The training data are derived from publicly available sources: GeoNames (<https://www.geonames.org/>), Wikidata (<https://www.wikidata.org/>), and the Getty Thesaurus of Geographic Names (<https://www.getty.edu/research/tools/vocabularies/tgn/>). The MEHDIE benchmark used for evaluation is available via the supplementary file link at <https://link.springer.com/article/10.1007/s10579-025-09812-9>.

Code Availability

Training code and inference utilities are available at:

<https://github.com/WorldHistoricalGazetteer/indexing>.

The repository includes technical documentation of the HDBSCAN clustering methodology used for positive pair generation. The system integrates with Elasticsearch for scalable deployment.

Statements and Declarations

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Competing Interests

The author has no relevant financial or non-financial interests to disclose.

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